Why crypto needs privacy now

Open ledgers made crypto verifiable; private networks will make it usable for business. Why privacy is becoming a condition for adoption, and what it means for UR on Bittensor SN25.

KHALA RESEARCH

Khala Research

November 2026, draft for internal review

Privacy is becoming a condition for crypto adoption, and UR puts Bittensor in the network layer

Crypto now carries enough ordinary economic activity that a public ledger has become a commercial barrier.

Encryption hides what was sent, but the connection still shows who sent it. SN25 turns that missing layer into a commodity that anyone can supply and the network prices in TAO.

What could break it

New enforcement against privacy developers, and fees that accrue to issuers and applications rather than network tokens.

Stablecoins now sit on ledgers anyone can read. Multicoin counted more than $310 billion in stablecoin supply and $121 billion in DeFi in February 2026. *p. 4*

That exposes companies, and AI makes it cheap to exploit. A month of onchain payments shows a competitor payroll, suppliers, customers and treasury moves. *p. 5–6*

The case for privacy is now commercial as well as ideological. Buterin and Durov reach the same conclusion, and US policy has moved from sanctions to stablecoin law. *p. 7–8*

Markets and companies are already paying for it. ZEC rerated in 2025 and a quarter of its supply is now shielded; Zama, Bron and Dfns report live payment and trading deployments. *p. 10–11*

All of it leaks at the network layer, where few projects work. Only UR and Nym, of the 12 projects we track, protect who is talking to whom. *p. 12–13*

UR's thesis works if operator fees replace emissions. SN25 prices routing capacity as a digital commodity; the test is whether privacy operators pay for it before emissions fall. *p. 15–19*

Sources: Multicoin Capital, 2026 investment thesis (Feb 2026); Zama and Bron; Dfns and Zama; a16z, Investing in Nym; UR, subnet design and implementation. Company-reported deployments are not independently verified.

Why privacy, why now

Stablecoins now sit on readable ledgers *4*

What a public ledger reveals *5*

AI raises the stakes *6*

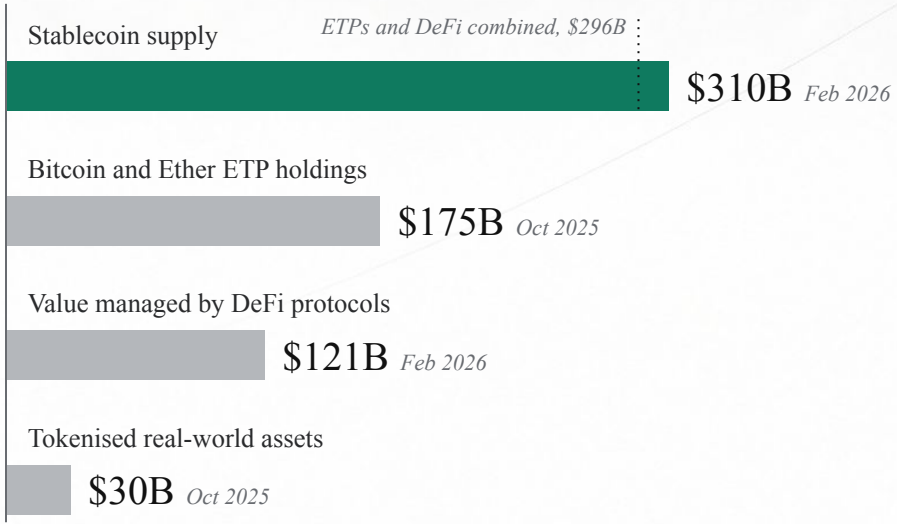
Two founders, one conclusion *7*

Policy has turned *8*

$310B of stablecoins now sits on ledgers anyone can read

Capital in selected onchain financial products, $B

Latest reported value; reporting dates differ by series



Evidence

Multicoin reported more than $310 billion in stablecoin supply and $121 billion managed by DeFi protocols in February 2026. a16z counted over $175 billion in Bitcoin and Ether exchange-traded products in October 2025.

What it means

Anyone who links an address to a company can read its payroll, supplier terms and positions. At this scale confidentiality becomes an operating requirement, which favours infrastructure that verifies transactions while limiting who can read them.

Qualification

These figures measure capital, not activity that needs privacy. Much stablecoin supply sits on exchanges, so the share that requires confidentiality is smaller and not yet measured.

Sources: Multicoin Capital, 2026 investment thesis (Feb 2026); a16z crypto, State of Crypto 2025 (Oct 2025). *Note:* Values as reported by each source, not adjusted for overlap between categories; the combined marker adds series with different reporting dates.

A month of onchain payments shows a competitor a company's payroll, suppliers and treasury moves

Fictional company, for illustration

Payment	*What the public ledger shows*	*What a competitor learns*	*What the public sees with controlled disclosure*
Salaries	Three transfers totalling $29,450 on the 25th of each month	Headcount and individual pay	Three payments of [redacted] to [redacted] ✓ *Verified by the network*
Supplier invoice	$184,000 to an address that analytics tools label Kestrel Components	The main supplier, its price and terms	One payment of [redacted] to [redacted] ✓ *Verified by the network*
Customer receipt	$420,000 in from an address labelled Orion Retail	A key customer and the contract size	One receipt of [redacted] from [redacted] ✓ *Verified by the network*
Treasury transfer	$2.4M to an exchange deposit address	A likely sale, before it settles	One transfer of [redacted] to [redacted] ✓ *Verified by the network*

What controlled disclosure means

The network still verifies every payment, but amounts and counterparties are readable only by the company, its auditor and regulators.

What it means

Businesses do not need secrecy, they need to choose who can read their records. Without that, putting payroll and treasury onchain publishes them.

Note: Fictional company, names and amounts. *Sources:* a16z, Investing in Seismic; Dfns and Zama, confidential institutional infrastructure.

AI makes public data easy to read and private data more valuable, so demand for private computation rises

Exposure gets cheaper

AI can analyse large volumes of public data. Transaction histories that once took real effort to interpret become easy to examine at scale.

Private data gets more valuable

Useful assistants need access to private communications, preferences and business records, so people and companies disclose more to them.

Demand for computation that uses data without exposing it

Nillion (Hack VC) builds private computation for AI and data. Seismic (a16z) computes over shared private data with secure hardware.

Sources: Vitalik Buterin, Why I support privacy (Apr 2025); Hack VC, Nillion investment thesis; a16z, Investing in Seismic.

Buterin and Durov argue from opposite ends but reach the same conclusion: privacy has to be the default

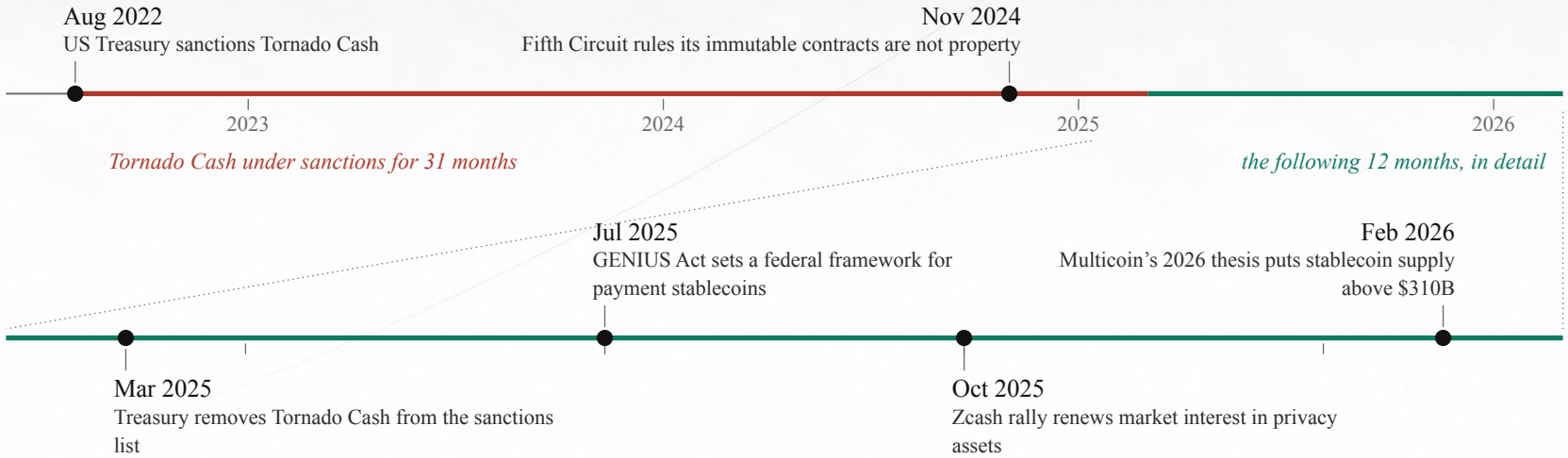
	Vitalik Buterin, Ethereum	Pavel Durov, Telegram
Why privacy matters	Control over information is power. Privacy protects decentralisation, individual freedom and institutions such as the secret ballot.	Encryption protects everyday communication, personal safety and freedom.
Where it should live	In the protocol: payments, application activity, blockchain queries and network anonymity, with Railgun and Privacy Pools in wallets.	In the product: no backdoors, because access built for authorities also exposes users to hackers.
How far they go	Selective disclosure, so data stays useful while misuse is limited.	Telegram would rather leave a market than add a backdoor.
The gap in practice	A roadmap; adoption depends on wallets and applications shipping it.	Telegram's default Cloud Chats are not end-to-end encrypted; only Secret Chats are.

Where they agree

Privacy has to be designed into the architecture rather than added later, which favours infrastructure built into wallets, chains and messaging over standalone tools.

Sources: Vitalik Buterin, Why I support privacy (Apr 2025); Vitalik Buterin, A maximally simple L1 privacy roadmap; Pavel Durov, statements on encryption and backdoors; Telegram, Cloud Chats and Secret Chats.

US policy has moved from sanctioning a privacy tool to legislating for stablecoins



Also in 2025, dates to confirm

Paxos and Aleo announce USAD, a private stablecoin. The Ethereum Foundation forms a dedicated privacy team.

What it means

The legal risk that kept institutions away has narrowed, and demand now comes from issuers, payroll providers and protocol teams.

Qualification

The ruling turned on whether immutable contracts are property. Developer liability remains open after a 2025 conviction.

Note: Upper axis to scale from Jun 2022; lower axis expands Feb 2025 to Mar 2026. *Sources:* US Treasury, OFAC (Aug 2022, Mar 2025); Fifth Circuit, Van Loon v. Treasury (Nov 2024); GENIUS Act (Jul 2025); Paxos; Aleo; Ethereum Foundation; Multicoin Capital (Feb 2026).

Where demand is forming

Zcash shows the market will pay *10*

Payments and trading lead *11*

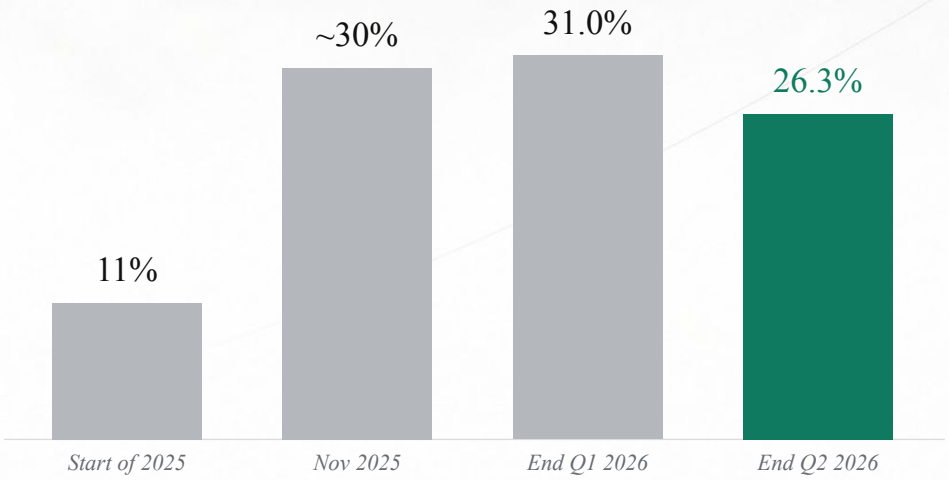
Where privacy still leaks *12*

Who protects the network layer *13*

Zcash’s rerating shows investors will pay for privacy, yet only about a quarter of ZEC is shielded

Share of ZEC supply held in shielded pools

Stock held, not share of transactions



Evidence

ZEC rose from about $50 in September 2025 to over $350 by late October, and Coin Metrics put it up nearly tenfold since October by mid-November. Shielded holdings reached about 4.9M ZEC.

What it means

Investors will pay for privacy, but most ZEC and most activity still sit on the transparent side: about $92,000 of $110,000 in Q2 2026 fee revenue came from transparent transfers. A private asset still leaks at its entry and exit points.

For UR

That leak sits at the connection, which is the layer SN25 supplies.

Note: Q1 and Q2 2026 figures come from a report funded by GenZcash; November 2025 is approximate. *Sources:* Coin Metrics, State of the Network 338 (18 Nov 2025); The Block (29 Oct 2025); Blockworks Advisory, Zcash token holder reports Q1 and Q2 2026.

Demand is most concrete in payments and trading, where companies already report live deployments

Use case	*What must stay private*	*Evidence so far*	*Who is building or backing it*
Payments and payroll *Company-reported deployments*	Salaries, supplier invoices and treasury movements	Zama and Bron report payroll on Ethereum using confidential USDT, with amounts encrypted	Zama, Bron, Aleo
Trading and tokenised assets *Company-reported deployments*	Orders, positions and holdings	Zama reports a confidential OTC trade involving GSR; Dfns has added confidential tokens to its wallet platform	Zama, Dfns *Backed by* Variant
Private computation and AI *Building*	Customer data and model inputs	Computation over shared private data with secure hardware, and private computation infrastructure	Seismic, Nillion *Backed by* a16z, Hack VC
Programmable privacy *Building*	Hidden inputs for sealed-bid auctions, private ballots and strategy games	Reusable tools, so each application does not need its own cryptography	Fhenix, Zama, Aztec, Noir *Backed by* Multicoin, a16z

What it means

Payments and trading lead because the cost of exposure is direct and easy to measure. Every row still reveals who is transacting, and from where, unless the connection itself is private.

Note: Deployments are company-reported and not independently verified. *Sources:* Zama and Bron; Aleo; Variant, Crypto privacy landscape; Zama, institutional applications; Dfns and Zama; a16z, Investing in Seismic; Hack VC; Multicoin, The dawn of onchain FHE; a16z, Investing in Aztec.

Encrypting a payment hides what was sent but not who sent it; only network privacy protects that

	Payments	*Computation*	*Applications*	*Network*
Amounts and balances	●	●	○	○
Counterparties	●	○	●	○
Inputs to computation and AI	○	●	○	○
Contract state and user activity	○	○	●	○
IP address and timing	○	○	○	●
Who talks to whom	○	○	○	●

Filled circle: protected by that layer. Rows reflect each layer's primary purpose.

What it means

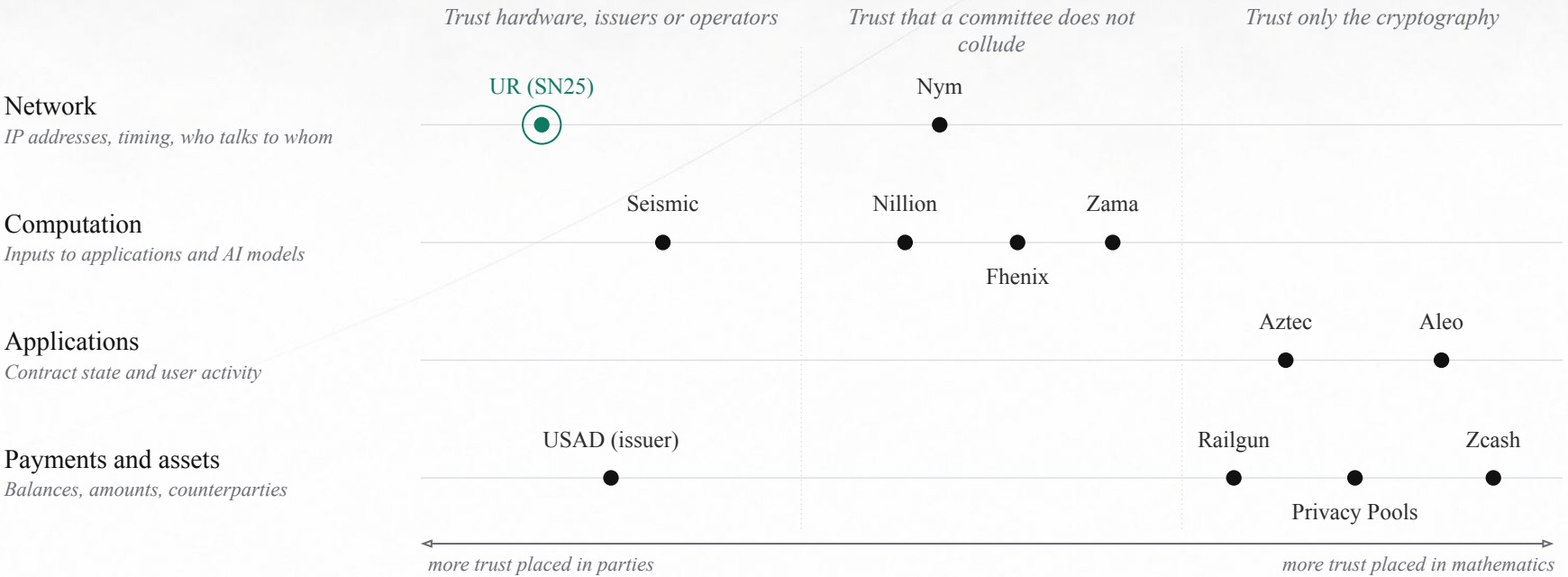
Payments, computation and application privacy protect the content of a transaction. The connection that carries it still shows an IP address, timing and who talks to whom.

Who works on it

Nym (a16z) runs an incentivised mixnet. Vitalik's roadmap adds network anonymity and private blockchain queries. UR supplies connectivity capacity.

Note: Simplified; several projects span layers. *Sources:* a16z, Investing in Nym; Vitalik Buterin, Ethereum privacy roadmap; UR, subnet design and implementation.

Of the 12 privacy projects we track, only UR and Nym protect network metadata

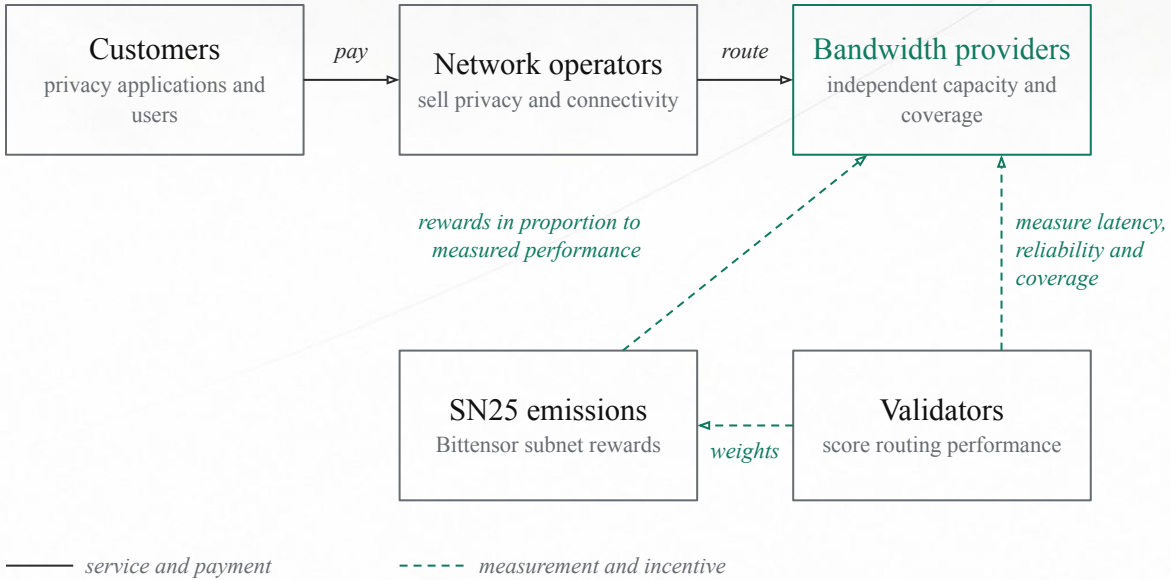


Note: Khala Research placement by primary trust assumption; several projects combine techniques and span layers. USAD is the private stablecoin announced by Paxos with Aleo. *Sources:* a16z crypto; Multicoin Capital; Hack VC; Vitalik Buterin, L1 privacy roadmap; Aleo.

The opportunity for UR

How SN25 works *15*
How UR should be positioned *16*
Sizing the fee pool *17*
Where value accrues *18*
The investment thesis *19*
What could break it *20*
Thoughts on Nock *21*
What we believe *22*

SN25 turns private bandwidth into a digital commodity that validators measure and TAO pays for



What the design provides

Reliable capacity and geographic coverage that many privacy operators can draw on, paid for through an incentive that rewards measured performance rather than claims.

What UR does not claim

UR does not claim protection against global traffic correlation and does not use mixing or cover traffic, so its positioning has to match these guarantees.

Why it matters

Network privacy is a separate problem from transaction privacy. Routing capacity of this kind complements mixnets such as Nym rather than replacing them.

Sources: UR, subnet design and implementation; URnetwork, how the network works; URnetwork, threat model. *Note:* Simplified for clarity; payment terms between operators and providers are not shown.

UR should be sold as capacity for privacy operators, not as a mixnet

	UR (Bittensor SN25)	Nym
Role in the stack	Connectivity capacity and geographic coverage	Mixnet that hides network metadata
What the buyer gets	Routing through independent bandwidth providers, sold by network operators	Access to a mixnet for apps and users
Protection against a global observer	Not claimed	Designed to resist traffic analysis through mixing and cover traffic
How supply is paid for	SN25 emissions, weighted by measured routing performance	Token rewards to mix node operators
Natural customer	Privacy and connectivity operators that need reliable capacity	Apps and users that need strong metadata privacy

What it means

UR competes on capacity, coverage and cost, and complements mixnets rather than replacing them. Marketing that implies mixnet-grade anonymity would be a reputational risk.

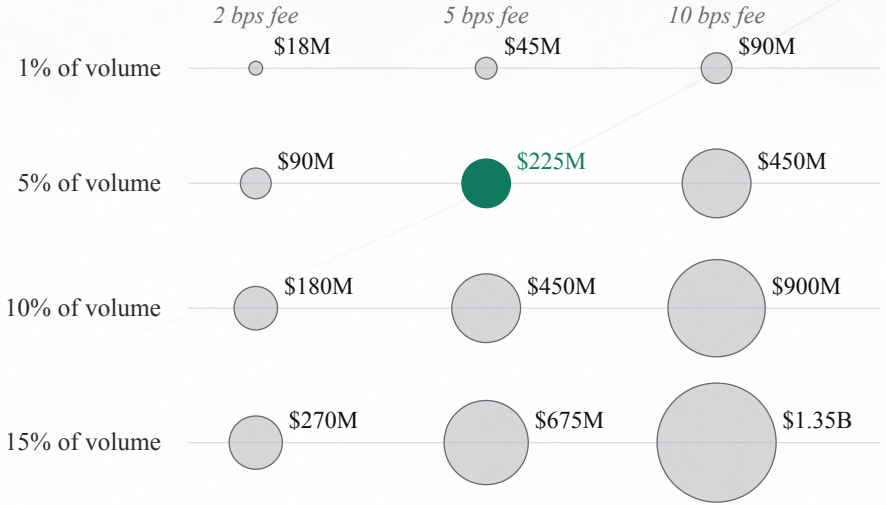
Note: Nym description simplified. *Sources:* UR, Subnet design and implementation; URnetwork, How the network works; URnetwork, Threat model; a16z, Investing in Nym.

Even 5% confidential adoption of stablecoin volume implies a fee pool of about $225M a year

Annual fee pool from confidential stablecoin payments

Illustrative data

$9T adjusted volume a year × confidential share × fee rate



Circle area is proportional to the annual fee pool. The base case is filled in jade.

Bear

$18M a year

1% of adjusted volume needs confidentiality, and competition between pools pushes fees to 2 bps.

Base

$225M a year

Payroll, treasury and business payments adopt confidential transfers: 5% of volume at 5 bps.

Bull

$1.35B a year

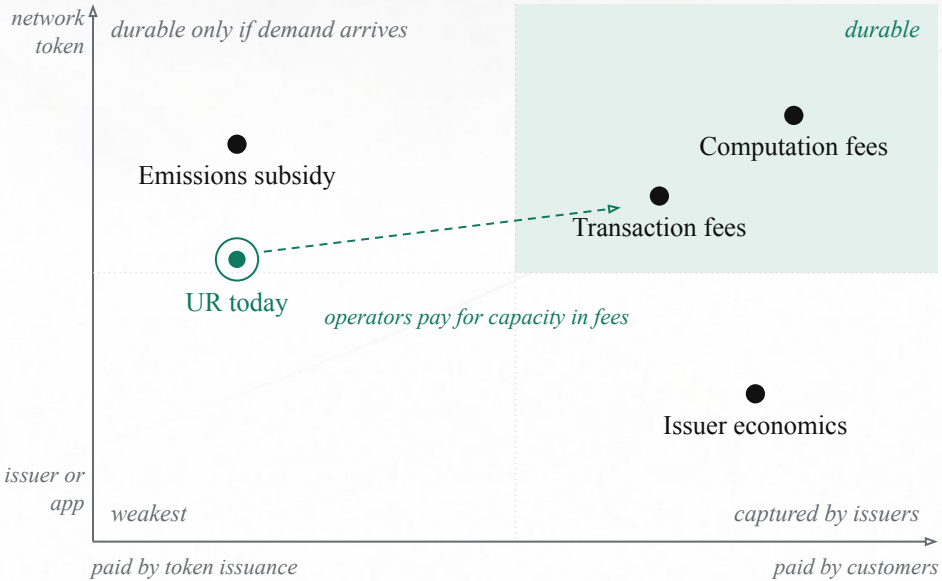
Regulated issuers make confidentiality a default feature: 15% of volume at 10 bps.

Qualification

The pool is shared across chains, pools, provers and issuers; volume is held at its 2025 level.

Note: Illustrative scenarios by Khala Research, not forecasts. 1 bp = 0.01%. *Source:* a16z crypto, State of Crypto 2025 (Oct 2025), adjusted stablecoin transaction volume over the prior 12 months.

UR sits in the weakest value-capture position today and has to move from emissions to operator fees



Computation fees. Applications and enterprises pay for encrypted operations, proofs and secure computation. Zama, Nillion, Seismic, Nockchain.

Transaction fees. Users pay per private transfer, shielding or withdrawal; pools compete on fees. Railgun, Zcash, Aleo.

Issuer economics. Reserve yield accrues to issuers and distributors, little reaches network tokens. Paxos USAD; Bron payroll.

Emissions subsidy. Token holders fund bandwidth and routing capacity through dilution before demand arrives. UR (SN25), Nym.

Implication for UR

The case depends on operators paying for UR's capacity in fees before subnet emissions decline, which would move it towards the transaction or computation models.

Note: Khala Research classification; positions are qualitative judgements of who pays and where value accrues, not forecasts. Fee mechanics vary by project. *Sources:* project documentation; Multicoin Capital; a16z crypto; Paxos and Aleo announcement.

The thesis holds if privacy demand turns into fees that customers pay repeatedly

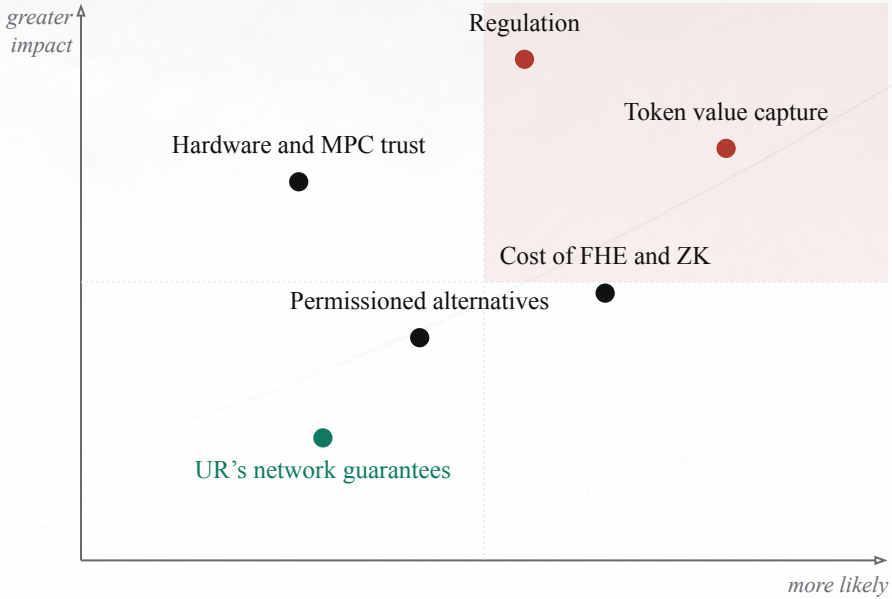
What has to be true	*Evidence today*	*What would prove it*
Privacy expands crypto's market	$310B in stablecoins; payroll and OTC deployments reported by Zama and Bron	Issuers and chains reporting a growing share of confidential volume
Products are usable	Security guarantees, cost and ease of use are still early	Confidential transfers available by default in mainstream wallets at costs close to public transfers
Privacy holds users	a16z argues users stay in private ecosystems because moving out risks leaking information	Users and balances retained in private environments over time
Demand is paid for repeatedly	Mostly pilots and token emissions today	Recurring fees; for UR, operator payments rising as SN25 emissions fall

What we will watch for UR

Operator payments against SN25 emissions, the number of paying operators, and coverage by region.

Sources: a16z, Privacy trends for 2026; Variant, Privacy infrastructure trade-offs; Multicoin Capital, 2026 investment thesis; Zama and Bron.

Regulation and token value capture are the two risks most likely to break the thesis



What would invalidate the thesis

Regulation. New enforcement against non-custodial privacy developers in the US or EU, after a 2025 conviction left developer liability unresolved.

Token value capture. Usage grows while protocol fee revenue stays flat, because fees accrue to issuers and applications.

Cost of FHE and ZK. Proving and FHE costs stop falling, or hardware acceleration is delayed.

Hardware and MPC trust. A material enclave exploit or collusion event at a major network.

Permissioned alternatives. Institutional stablecoin volume concentrates on permissioned ledgers.

UR's network guarantees. Privacy operators require mixing or cover traffic that UR does not offer.

Note: Khala Research's qualitative judgement of likelihood and impact; provisional. The shaded quadrant holds the risks that matter most. *Sources:* US v. Storm, SDNY (Aug 2025); URnetwork, threat model; Vitalik Buterin, L1 privacy roadmap; Multicoin Capital.

Thoughts on Nock: Nockchain pays miners to produce zero-knowledge proofs

$7.5M

Reported funding, including a $5M seed round led by Delphi Ventures

Developer	Zorp
Token	NOCK, newly issued to miners
Consensus	Proof-of-work in which miners generate zero-knowledge proofs
Other investors	North Island Ventures, CMCC Global and others
DCG	No direct investment verified

How it works

Bitcoin miners compete on hash puzzles with no use outside the network. Nockchain miners instead compete to produce zero-knowledge proofs, so the work that secures the chain is the kind of computation applications could pay for. A proof lets a party verify a computation without seeing its inputs, which extends privacy into financial services and confidential computing.

Investment thesis

Investor commentary places NOCK alongside alternative stores of value, new proof-of-work coins and privacy coins, combining Bitcoin's monetary properties with programmable public and private applications.

What has to be true

Demand emerges for the proofs miners produce, beyond the work needed to secure the chain.

Proof generation on Nockchain is cost-competitive with dedicated proving services.

Enterprise users accept a proof-of-work network for confidential computation.

Open questions

The same commentary linked NOCK's October 2025 price rise to the Zcash rally, so recent performance may reflect sentiment toward privacy coins more than usage. Funding details are as reported by the company and investors.

Sources: Zorp and Nockchain project materials; Delphi Digital, $NOCK (13 Oct 2025); investor commentary attributed to Zurlo in draft notes (attribution to be confirmed).

Privacy should be built into the network, supplied by anyone and paid for by the people who use it

Private by default

Payments, queries and connections should reveal only what their owners choose to share. Disclosure should be something a user decides to make, with an auditor or a regulator, and never a side effect of using the network.

Open supply

Anyone with bandwidth, hardware or code should be able to add capacity from wherever they are. A network that depends on one operator, or one country, can be switched off.

Measured, then paid

Validators should measure what each provider actually delivers, and rewards should follow that measurement. Over time those rewards should come from customers rather than from emissions.

Bittensor showed that an open network can produce intelligence and price it as a commodity. Privacy is the next commodity that belongs on such a network, and SN25 is where it can be supplied, measured and paid for.

Note: Khala Research view. *Sources:* Bittensor, The Bittensor Paradigm; UR, subnet design and implementation.

How this report was prepared, and the sources behind it

Methodology

Scope. Privacy infrastructure for public blockchains across four layers: network, computation, applications, and payments and assets.

Evidence. Public investor research, project documentation, court records and regulatory releases. Figures are as published and not adjusted for overlap.

Illustrative data. Modelled or hypothetical figures carry a sand Illustrative data tag and are not forecasts.

Assessments. Qualitative ratings and placements are Khala Research judgements and are labelled as such.

Disclosures

[Placeholder] Holdings, conflicts of interest and compensation to be confirmed before publication. This draft is for internal review and is not investment advice.

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Aleo, Paxos Labs and the Aleo Network Foundation launch USAD

Bittensor, The Bittensor Paradigm

Note: Draft reference list. Confirm titles, dates and links against the originals before publication.

Appendix: privacy projects compared by what they protect and whom users must trust

Project and cited by	*What it protects*	*Technique*	*Whom users must trust*	*Reported traction*
Zama *Multicoin*	Balances, amounts and contract state	Fully homomorphic encryption	Cryptography; a threshold key network decrypts	Payroll with Bron; OTC trade with GSR (company-reported)
Aztec *a16z*	Private application logic and state	Zero-knowledge rollup	Cryptographic proofs	*n/r*
Seismic *a16z*	Computation over shared private data	Secure hardware (TEE)	Hardware vendor and enclave integrity	*n/r*
Nillion *Hack VC*	Private computation for AI and data	Multi-party computation	Non-collusion among node operators	*n/r*
Nym *a16z*	Network metadata: who talks to whom	Incentivised mixnet	Non-collusion among mix nodes	*n/r*
Railgun, Privacy Pools *Buterin roadmap*	Payment amounts and counterparties	Zero-knowledge shielded pools	Proofs; size of the anonymity set	Railgun: over $200M a month (a16z)
UR (Bittensor SN25) *Subject of this report*	Connectivity capacity for privacy services	Incentivised bandwidth marketplace	Operators; no global-correlation defence	Subnet design documented

Note: n/r = not reported in our sources. Trust assumptions are simplified. Tinted row marks the subject of this report. *Sources:* a16z crypto investment posts; Multicoin Capital; Hack VC; Vitalik Buterin, L1 privacy roadmap; a16z, State of Crypto 2025; UR subnet design documentation.